

Project Design Phase
Problem – Solution Fit Template

Date	30 September 2026
Team ID	SWTID-2026-9682
Project Name	AI StudyBuddy – AI-Augmented Backend Development
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ☐ Solve complex problems in a way that fits the state of your customers.
- ☐ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ☐ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ☐ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ☐ **Understand the existing situation in order to improve it for your target group.**

Template:

1. CUSTOMER SEGMENT(S) [CS] • Engineering / college students preparing for semester exams, internships and placement tests (e.g., Rahul, a 2nd-year student). • Competitive-exam aspirants. • Administrators who manage the platform.	6. CUSTOMER CONSTRAINTS [CC] • Very limited time before exams. • Low or no budget for paid tools. • Study material scattered across PDFs, books, notes and web articles. • Needs an internet connection and a device for AI features. • Concern about accuracy of AI output.	5. AVAILABLE SOLUTIONS [AS] • Pen, paper and highlighters: free but slow and repetitive. • Flashcard apps (Anki, Quizlet): good recall, but cards are made manually. • Note apps (Notion, Google Keep): store notes, no AI generation. • General chatbots: need prompting; results are not saved or organised per subject.
2. JOBS-TO-BE-DONE / PROBLEMS [J&P] • Understand lengthy material quickly. • Make concise revision notes. • Create flashcards and practice questions. • Test knowledge with quizzes. • Plan revision around exam dates. • Keep all resources in one secure place.	9. PROBLEM ROOT CAUSE [RC] • Large syllabus and short preparation time. • Note-making and question-making are manual and repetitive. • Tools are fragmented (notes, quizzes, planner in separate apps). • No personalisation to the student's exam schedule.	7. BEHAVIOUR [BE] • Highlight and copy notes into documents. • Search YouTube and articles for short explanations. • Ask chatbots ad hoc, without saving results. • Cram at the last minute. • Switch between several apps to study.
3. TRIGGERS [TR] • Exam timetable is announced. • Internship or placement test approaching. • Receiving a large PDF or lecture deck. • Low score in a mock test.	10. YOUR SOLUTION [SL] • AI StudyBuddy: a secure REST backend (Node.js, Express.js, MongoDB) integrated with the Google Gemini API. • A student uploads or pastes material and gets an AI summary, flashcards, an MCQ quiz and a personalised study plan. • JWT authentication, bcrypt password hashing and RBAC protect user data. • All generated resources are stored for later revision; admins monitor users and usage.	8. CHANNELS of BEHAVIOUR [CH] • 8.1 ONLINE: web dashboard, YouTube, online PDFs and articles, WhatsApp/Telegram study groups, college portals. • 8.2 OFFLINE: lecture notes, textbooks, handwritten notes, library, peer discussions, printed timetables.
4. EMOTIONS: BEFORE / AFTER [EM] • Before: overwhelmed, anxious, unsure what is important. • After: confident, organised and in control of revision.		

Canvas filled for AI StudyBuddy (numbers 1–10 follow the Problem–Solution Fit canvas order).

References:

Problem–Solution Fit Canvas (Idea Hackers): <https://www.ideahackers.network/problem-solution-fit-canvas/>

Problem-Solution Fit Canvas (Medium): <https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe>

Google Gemini API Documentation: <https://ai.google.dev/gemini-api/docs>

Anki (spaced-repetition flashcards) – existing solution: <https://apps.ankiweb.net/>

Quizlet – existing solution: <https://quizlet.com/>